Deployment preparation Build v0.4

Django + durable PostgreSQL, prepared for the selected later GitHub-to-Vercel route.

This is tested local source, not a hosted client release. No repository was pushed, account created, plan purchased or deployment published. October 26 at 5 PM IST remains the target, not a guarantee.

What is ready in the code

– Existing employee/joining/salary records, work allocation, CRM enquiries and Maintenance jobs remain. Technicians can update only their assigned service jobs; salary, CRM and dispatch restrictions remain.

– TLS PostgreSQL support, database-backed sessions, strict Vercel production settings and explicit migration/release tooling. Local SQLite development remains available.

– Foreground Start/Stop GPS is preserved. Dispatcher now removes expired displayed positions without a refresh while JavaScript runs. A protected cleanup endpoint is ready, but no recurring cloud schedule is installed.

Local verification

Check	Result and boundary
135 automated tests	Passed on SQLite and local TLS-enabled PostgreSQL.
14 browser workflow checks	Passed in Chromium with fictional records and simulated GPS; not physical Android.
3 expiry checks	Stale label and removal confirmed using accelerated browser clock.
8 PostgreSQL rehearsal checks	Migration retention, independent-process sessions/data, release plan/apply, restore and TLS passed.
Production/recovery rehearsal	Local Gunicorn + TLS and SQLite backup/recovery passed. No hosted recovery tested.

Raw results and full operator instructions are included in the source archive. Prior version evidence is retained as history, not a current production certificate.

Use an existing suitable paid team first

Keep Django; Vercel now officially supports it. Use external managed PostgreSQL, not SQLite on a serverless filesystem. First inspect the owner's existing Vercel team plan: a suitable active Pro or Enterprise team may already cover this project. The account plan has not been inspected; no new subscription is assumed.

Vercel Hobby is restricted to non-commercial personal use. This commercial client system should not be hosted there. Current Pro pricing is USD 20/month platform fee, including one deploying seat and USD 20 usage credit. Extra deploying seats are USD 20/month each; overage, optional features and applicable taxes can add cost. Confirm existing entitlements, spend controls and a budget before approval.

A concrete PostgreSQL option

Neon is a candidate, not a purchased provider. Use the pooled URL for app traffic and a direct URL for migrations/backups. Align Vercel compute with the DB region. Neon documents Singapore and Vercel supports sin1; confirm that region and cross-border data handling are acceptable and available to the account. No region is fixed in this source configuration.

Neon Launch currently has no monthly minimum: USD 0.106/CU-hour and USD 0.35/GB-month storage, plus other applicable usage/restore charges. Free currently includes 100 CU-hours/project and 1 GB/project; those limits do not establish production suitability.

Important cost interaction: a one-minute cleanup query can keep PostgreSQL awake. At 0.25 CU for 730 hours, compute alone is approximately USD 19.35/month; storage, restore history, transfer and tax are extra. That pattern exceeds 100 free CU-hours. This is a scenario, not an all-in quote or a spending authorization.

Before selecting or creating anything

– Exact private repository/owner and Vercel team/project; permission to push and deploy later.

– Existing paid-plan status, or separately approved spending if an upgrade/additional usage is needed.

– Database provider/region, usage limits and restore retention; domain/approved exact hostname.

All published prices/limits were checked against official pages on 19 October 2026 and must be refreshed before commitment. No trial with future charges should be activated without approval.

Configure privately; migrate once

Set protected production environment values. Preview needs a separate database, secrets and fictional records. Do not paste secret values into chat or commit them to GitHub. The example env file is a reference, not automatically loaded.

Setting	Required decision/value
DJANGO_SETTINGS_MODULE	config.vercel
DJANGO_DEBUG	0
DJANGO_SECRET_KEY	Unique high-entropy value, at least 50 characters
DJANGO_ALLOWED_HOSTS	Exact approved hostnames; no wildcard, scheme or port
DJANGO_CSRF_TRUSTED_ORIGINS	Exact HTTPS origins
DATABASE_URL	Pooled external PostgreSQL URL with TLS
CRON_SECRET	Separate random value, 32+ characters, for approved cleanup

The DB parser defaults to verify-full, supports sslmode=require where appropriate, and rejects insecure production fallback. URL-encode credentials. Provider CA/authentication, pooler behavior and actual network limits still need hosted testing.

With the direct DB URL loaded privately, VERCEL=1 and the production settings, first review the plan:

```
uv run --frozen --no-dev python scripts/release_database.py --confirm-host EXACT_DIRECT_HOST
--confirm-database EXACT_DB_NAME
```

After a tested backup/recovery point and single-operator signoff, repeat with --apply. This creates schema and shared throttle cache, not a deployment. Do not migrate during builds or each request; never use scripts/manage.sh against a hosted database because it deliberately selects local development.

Provision the first administrator privately with manage.py createsuperuser, then named technician mappings. No default passwords, activation mail or automatic signup is supplied. Restore the pooled app URL in Vercel.

After publication approval, connect the selected repository/project. Django detection handles WSGI and automatic collectstatic. Keep the app root at manage.py; no migration Build Command. Actual Vercel cloud build/bundle and runtime are not yet verified. Full commands and references are in docs/VERCEL_PREPARATION_v0.4.md.

Foreground consent is not background tracking

Location starts only after a technician taps Start and gives browser permission. Stop revokes the session. Hiding/leaving the page stops its watch and attempts server deletion; a lost connection can prevent immediate server deletion.

Only the latest sample is kept, not a route trail. It becomes stale after 30 seconds and is excluded from views within 120 seconds without an accepted update. v0.4 removes expired rendered cards while its JavaScript runs and rechecks when visible. Suspended/browser-disabled code cannot provide a real-time guarantee. Refresh is still needed to fetch new positions.

The protected GET endpoint /internal/cron/purge-locations/ deletes expired rows using a separate CRON_SECRET bearer token. No schedule is installed. After plan/privacy approval, a Pro/Enterprise one-minute schedule is a candidate; verify actual execution and alerting. Hobby Cron only runs once daily, within the chosen hour. No cloud schedule guarantees exact deletion.

Provider backups, logs and point-in-time recovery can still retain deleted samples. Agree and verify those retention rules separately. An application TTL is not proof of erasure from all storage.

Backup and restore deliberately

The local PostgreSQL rehearsal restored five business tables and administrator authentication into a separate database, excluding location/session/throttle rows. Production backup storage, encryption, retention and actual provider restore remain open.

The full guide includes a pg_dump example that excludes operations_locationshare, django_session and login_throttle data. Use direct PostgreSQL access and protected authentication; encrypt the dump and restore to a new database before relying on it. The SQLite snapshot script is not a PostgreSQL backup tool.

Must still be checked on real Android + HTTPS

– Permission allow/deny, Start/Stop, account revocation, network loss and tab visibility/locked-phone lifecycle.

– Admin dispatcher expiry and authorized technician-only job access on the actual host.

Browser GPS cannot promise reliable locked/background updates, attendance proof or safety-critical tracking. Coordinates in tests were simulated; no actual person’s location was read.

What remains before client handover

– Agree the role matrix: ordinary operators currently share generic allocations. Confirm operational timezone; timestamps remain UTC, not an inferred client timezone.

– Settle actual administrator/backup accounts, onboarding/password recovery and training. Private provisioning and password changes work; activation/recovery email is not implemented or verified.

– Test hosted database migrations, persistence across deployments, pooler/TLS, backup/restore, secrets, exact hosts, CSRF, static files, logs and cleanup during inactivity.

– Complete real Android/laptop review, realistic concurrency/load and independent security review. Local tests alone cannot certify production safety.

– Approve privacy/retention, encrypted off-host recovery, downtime/rollback instructions, named operator and final client acceptance.

The original August baseline, quotation details and reliable voice-note content remain unavailable. Current CRM/Maintenance fields implement broad owner-accepted flows, not a complete client-approved contract. No inventory stock movements, invoices, payroll, automated notifications, route history or background tracker has been silently added.

Local startup

```
./scripts/setup-local.sh
./scripts/manage.sh createsuperuser
./scripts/manage.sh runserver 127.0.0.1:8000
```

Extract the source and run from its powerline-v0.4 folder. Python/uv required. Enter passwords privately; use fictional local data and loopback access. Current full operating guide, role mapping, migration commands and evidence are inside the ZIP.

Official references

https://vercel.com/docs/frameworks/full-stack/django

https://vercel.com/docs/plans/hobby

https://vercel.com/docs/plans/pro-plan

https://vercel.com/docs/cron-jobs/usage-and-pricing

https://neon.com/docs/guides/django

https://neon.com/docs/guides/django-migrations

https://neon.com/pricing

More official region, runtime and serverless-pooling references are in the source guide. Plan entitlements and current prices must be checked again before any commitment.